

Gaussian & Bilateral Filtering

Analysis Report

This project is a task submission for the Digital Image Processing course of Computer Science undergraduate in Universitas Gadjah Mada. It focuses on implementing gaussian filtering and bilateral filtering on high-noise images and comparing the results of both algorithms.

Gaussian Filtering:

Before:



After:



(kernel size = 5, $\sigma = 5$)

Gaussian filtering is a linear filter that works by blurring the whole image indiscriminately. It weighs neighboring pixels based only on their spatial distance from the center pixel. This smooths out noise but also blurs out important sharp edges, which produces an image with less noticeable noise, albeit with less preserved details.

Bilateral Filtering:

Before:



After:



(kernel size = 5, $\sigma_s = 5$, $\sigma_r = 25$)

Bilateral filtering is a non-linear filter that works by combining spatial distance with intensity differences. It weights pixels based on both spatial distance (domain) and intensity/color similarity (range). This means it preserves edges by ignoring pixels with significantly different intensity values, which creates an image with less noise while still maintaining important edges and details.